

**SONA COLLEGE OF TECHNOLOGY**  
**(AUTONOMOUS INSTITUTION)**  
**DEPARTMENT OF INFORMATION TECHNOLOGY**  
**MINI PROJECT – U23IT705 (2026-27)**

**DATE: 29.07.2026**

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|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>BATCH NO:</b>                                     | 10                                                                                                                                                                                                                                                                                                                   |                  |
| <b>REGISTRATION NUMBER</b>                           | <b>STUDENT NAME</b>                                                                                                                                                                                                                                                                                                  | <b>SIGNATURE</b> |
| 61782323106044                                       | JAYAKUMAR M                                                                                                                                                                                                                                                                                                          |                  |
| 61782323106072                                       | NEHA PRAPTI P                                                                                                                                                                                                                                                                                                        |                  |
| 61782323106077                                       | PRABANJAN V                                                                                                                                                                                                                                                                                                          |                  |
| <b>TITLE OF THE PROJECT:</b>                         | CLOUDLAB INTELLIGENCE                                                                                                                                                                                                                                                                                                |                  |
| <b>DOMAIN NAME/AREA:</b>                             | CLOUD COMPUTING                                                                                                                                                                                                                                                                                                      |                  |
| <b>Algorithm/Technique used:</b>                     | Activity Duration Calculation Algorithm, Idempotent Data Synchronization, Context-Switch Analysis                                                                                                                                                                                                                    |                  |
| <b>REFERENCE (JOURNAL/ ONLINE RESOURCES/ OTHERS)</b> | <a href="https://aws.amazon.com/what-is/cloud-computing/">https://aws.amazon.com/what-is/cloud-computing/</a><br><a href="https://martinfowler.com/articles/patterns-of-distributed-systems/idempotent-receiver.html">https://martinfowler.com/articles/patterns-of-distributed-systems/idempotent-receiver.html</a> |                  |
| <b>GUIDE NAME &amp; SIGNATURE</b>                    | <b>MR. P. DINESHKUMAR,</b><br><b>AP/ IT</b>                                                                                                                                                                                                                                                                          |                  |

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**ABSTRACT**

**CloudLab Intelligence: An Offline-First Edge-Cloud Platform for Student Digital Activity Analytics and Predictive Computer Lab Resource Optimization** — College computer laboratories often lack a unified system to accurately monitor student sessions, computer utilization, application usage, browser activity, and resource requirements. Existing cloud-dependent systems may also become unavailable during internet outages, resulting in incomplete activity records. To address these challenges, CloudLab Intelligence proposes an offline-first Edge-Cloud platform that continuously records authorized student and computer activity. Students authenticate using their register numbers, and each session is associated with the student, computer, laboratory, and accurate timestamps. A lightweight monitoring agent records permitted metadata such as login/logout events, application usage, active browser domains or tabs, tab-switch events, and time spent on different activities. A local edge server stores these events in an offline database and continues operation without internet connectivity. Once connectivity is restored, the stored data is automatically synchronized with the cloud. The system uses a Hybrid Edge-Cloud Architecture consisting of monitoring agents, a local edge server, synchronization module, cloud backend, analytics engine, and dashboard. The Event-Based Logging Algorithm records activities with unique IDs and timestamps, while the Session Tracking and Activity Duration Calculation Algorithms determine session and activity durations. The Offline-First Synchronization Algorithm manages local storage and cloud synchronization during network failures, supported by an Idempotent Synchronization Algorithm to prevent duplicate records. A Rule-Based Activity Classification Algorithm categorizes activities into Programming, Database, Research, Learning, and Documentation, while a Context-Switch Analysis Algorithm evaluates application and browser transitions. Machine-learning algorithms such as Linear Regression, Random Forest Regression, or XGBoost can predict laboratory demand and resource utilization using historical data. Thus, the proposed system integrates Cloud Computing, Edge Computing, Offline-First Architecture, Activity Analytics, and Machine Learning to provide reliable monitoring, intelligent laboratory analytics, and predictive resource optimization for efficient computer laboratory management.

**PROJECT COORDINATOR**